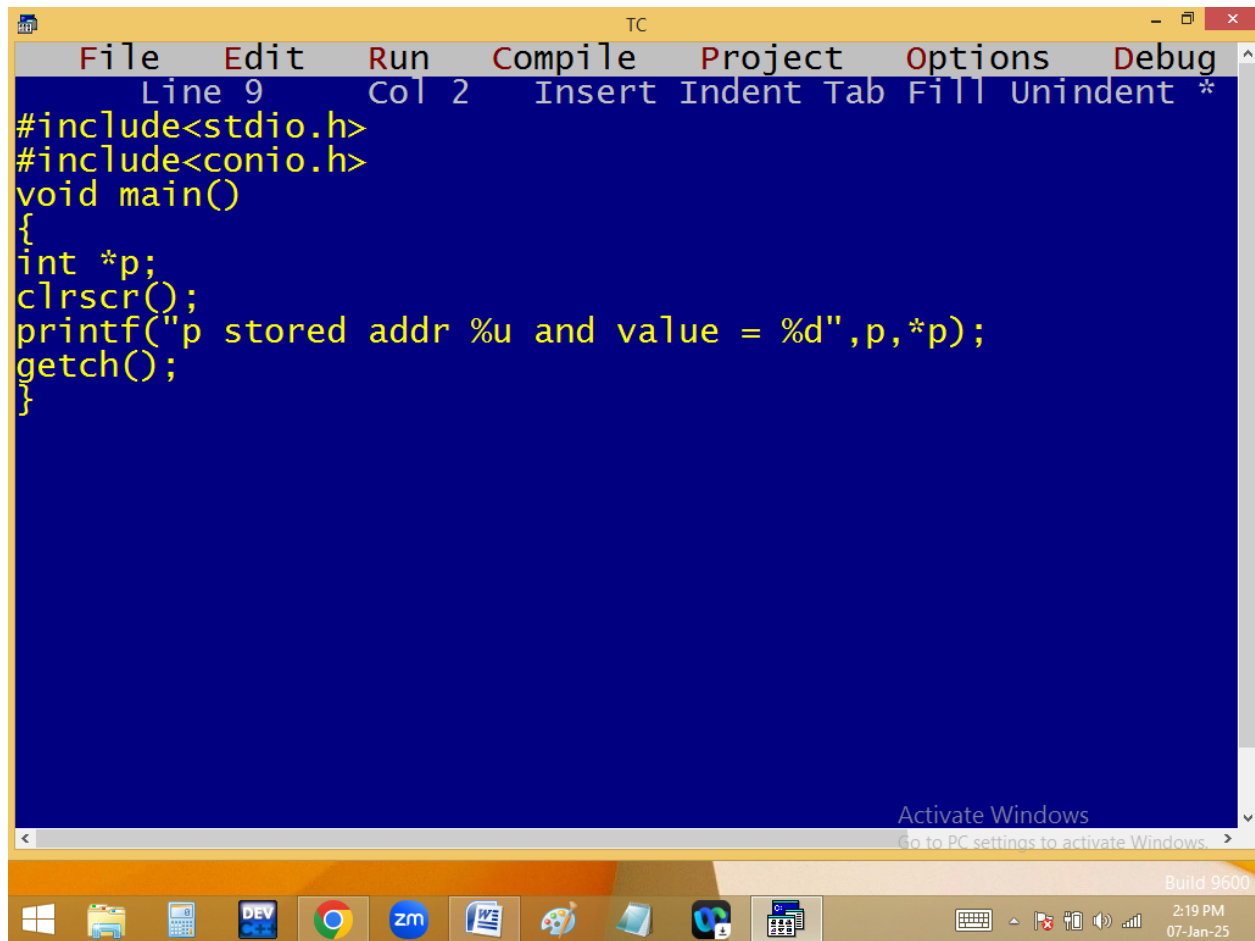


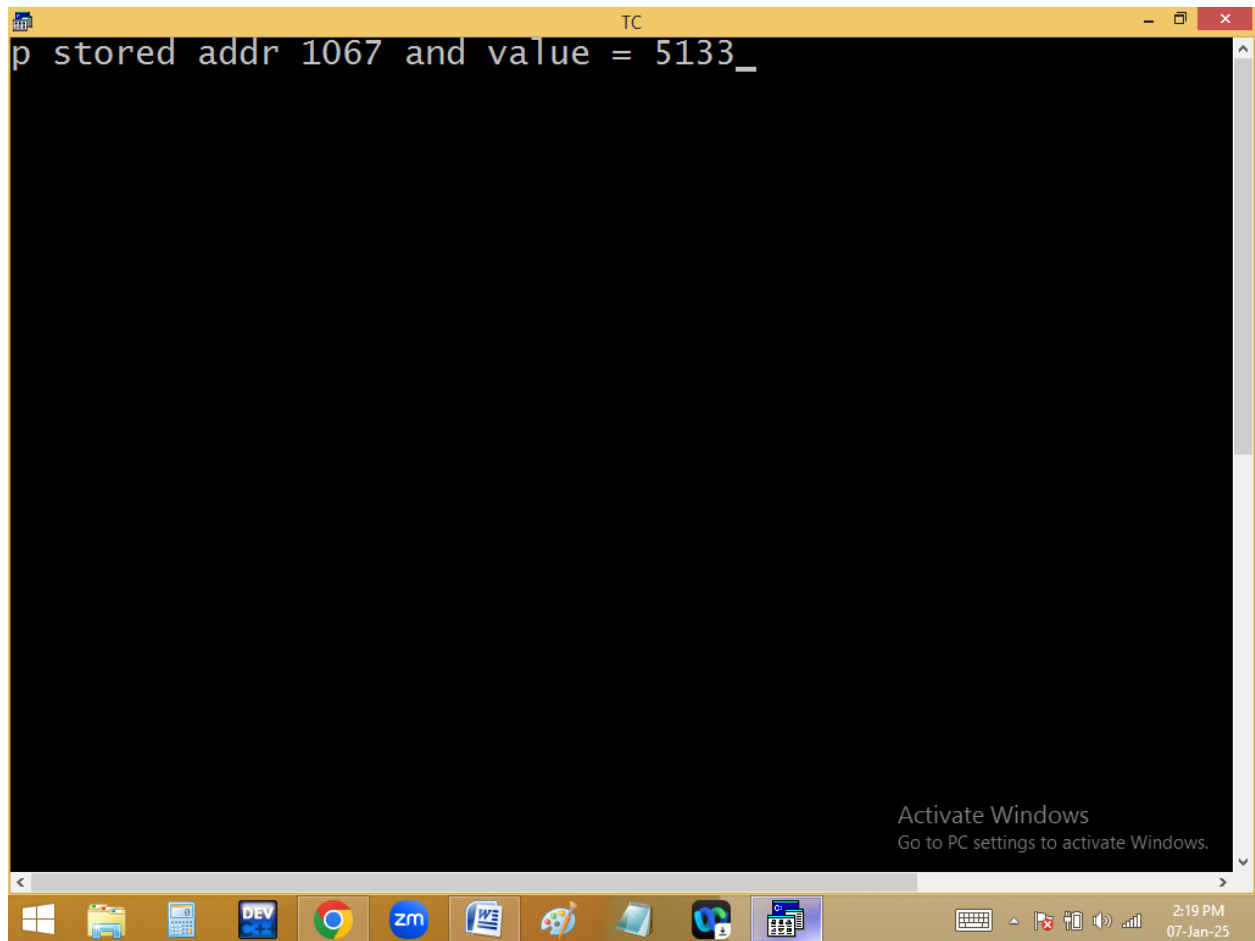
Bad / wild pointer: A pointer is declared but not initialized. In this situation the pointer is taking some unknown address and value. This kind of pointer is called bad / wild pointer.



```
TC
File Edit Run Compile Project Options Debug
Line 9 Col 2 Insert Indent Tab Fill Unindent *
#include<stdio.h>
#include<conio.h>
void main()
{
int *p;
clrscr();
printf("p stored addr %u and value = %d",p,*p);
getch();
}
```

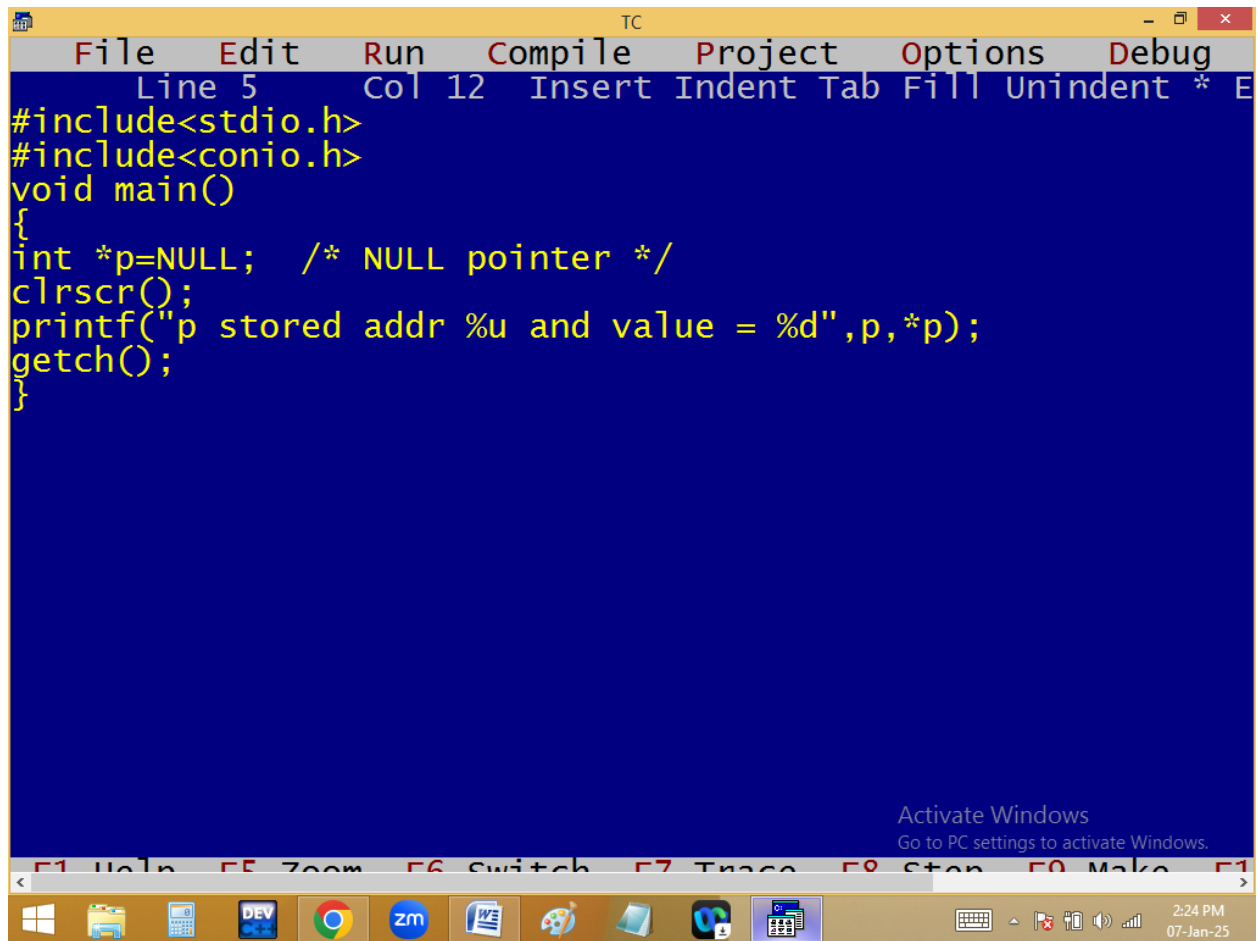
Activate Windows
Go to PC settings to activate Windows.

Build 9600
2:19 PM
07-Jan-25



```
p stored addr 1067 and value = 5133_
```

NULL pointer: When a pointer initialized with NULL /0 then it is a NULL pointer. To avoid bad and dangling pointers we are using NULL pointer.



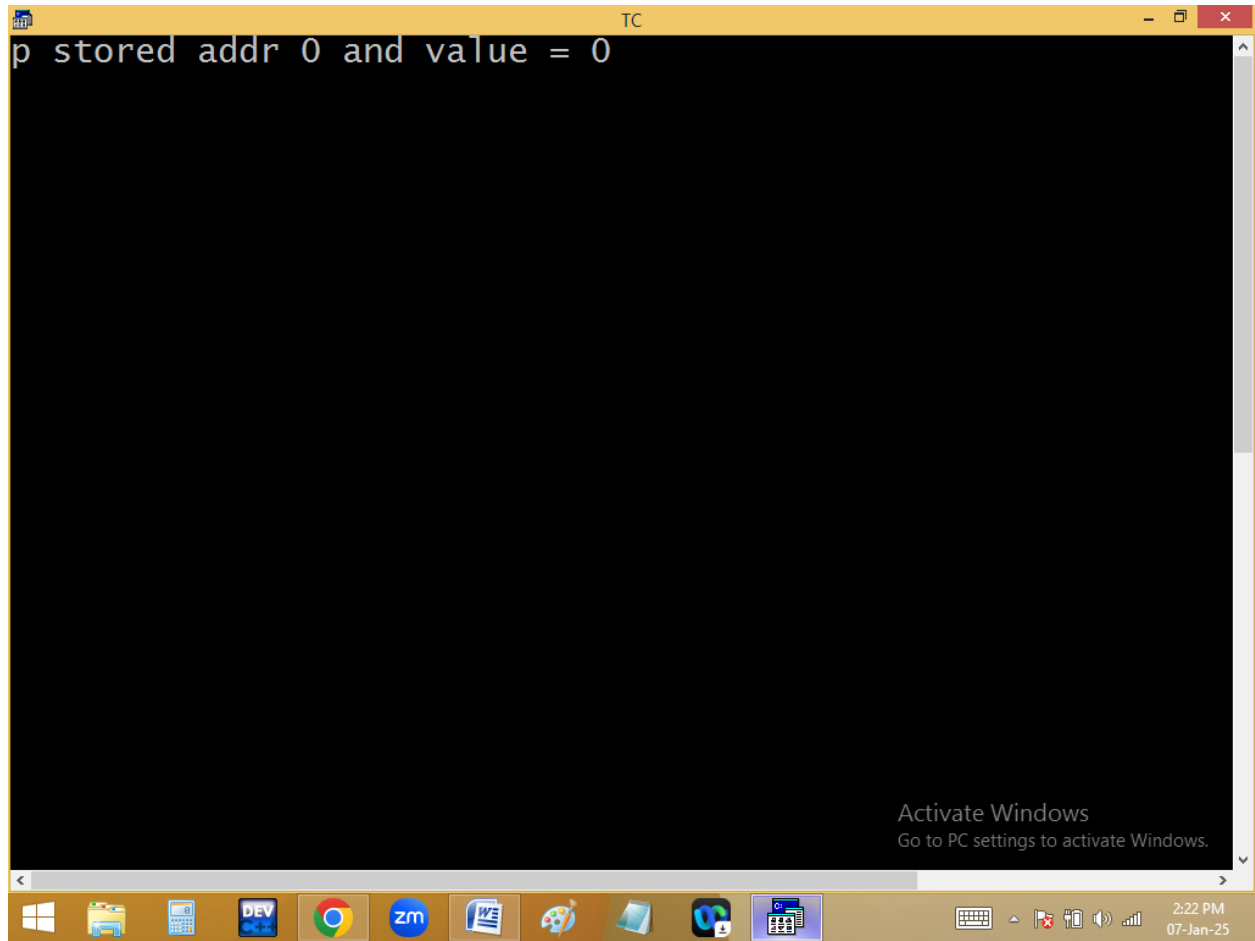
```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 12 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *p=NULL; /* NULL pointer */
clrscr();
printf("p stored addr %u and value = %d",p,*p);
getch();
}
```

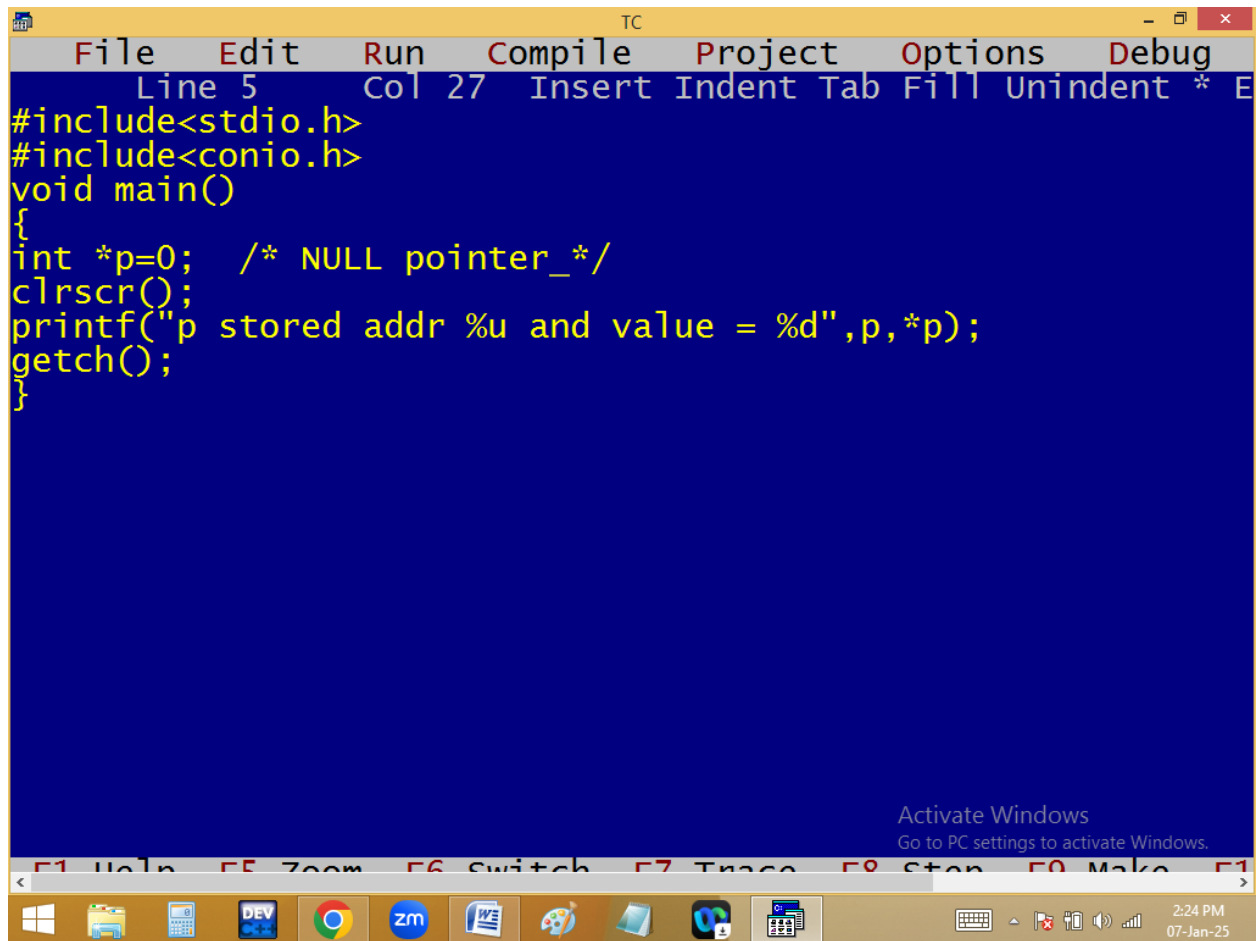
Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: File Explorer, Calculator, DEV C++, Chrome, Zoom, Word, Paint, VS Code, Task Manager.

System tray: Keyboard, Network, Volume, Signal, 2:24 PM, 07-Jan-25





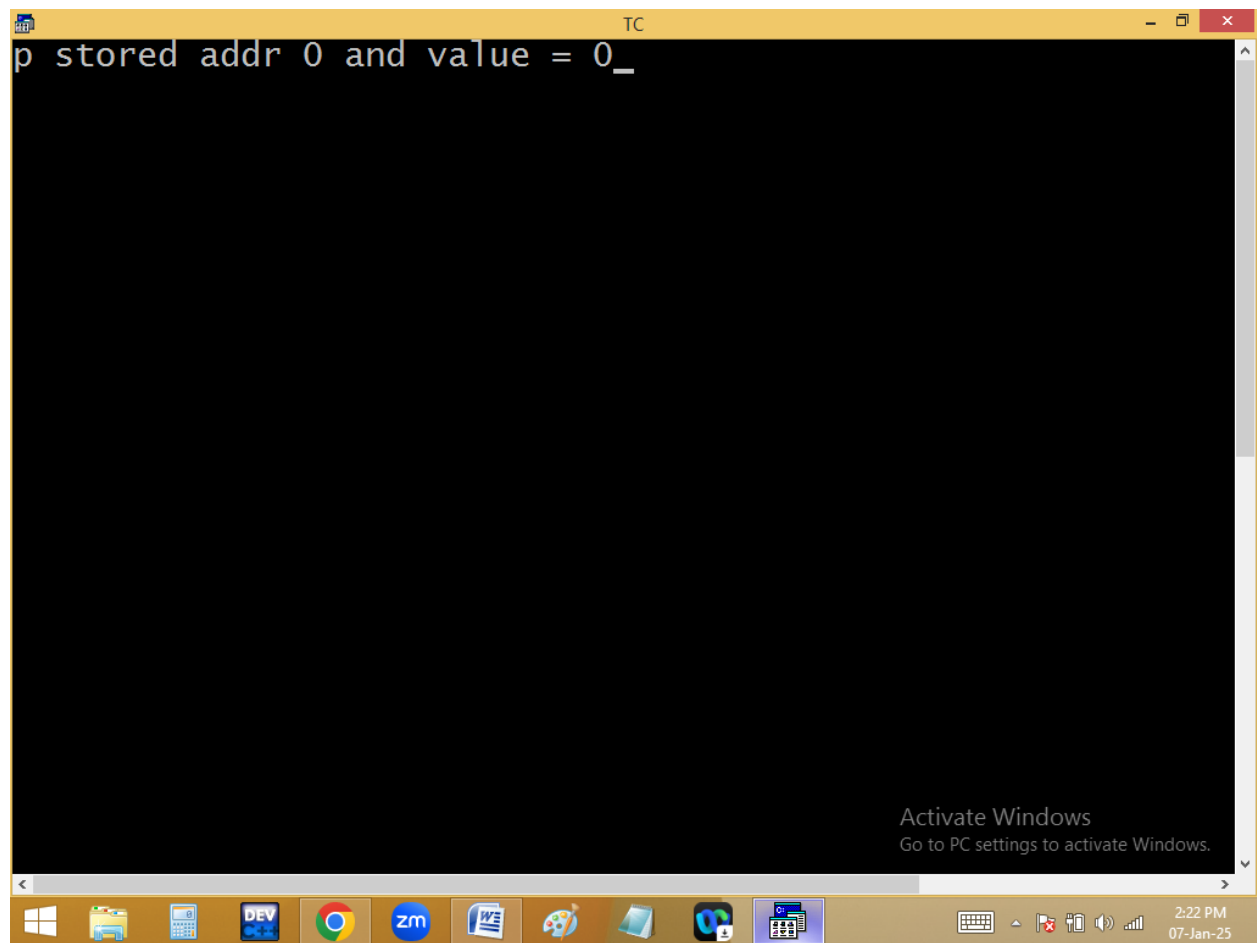
```
TC
File Edit Run Compile Project Options Debug
Line 5 Col 27 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *p=0; /* NULL pointer_*/
clrscr();
printf("p stored addr %u and value = %d",p,*p);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: File Explorer, Calculator, DEV C++, Chrome, Zoom, Word, Paint, VS Code, Task Manager.

System tray: Keyboard, Mouse, Network, Volume, Date/Time: 2:24 PM 07-Jan-25



Dangling pointer:

TC

File Edit Run Compile Project Options Debug

Error: Undefined symbol 'a' in function main

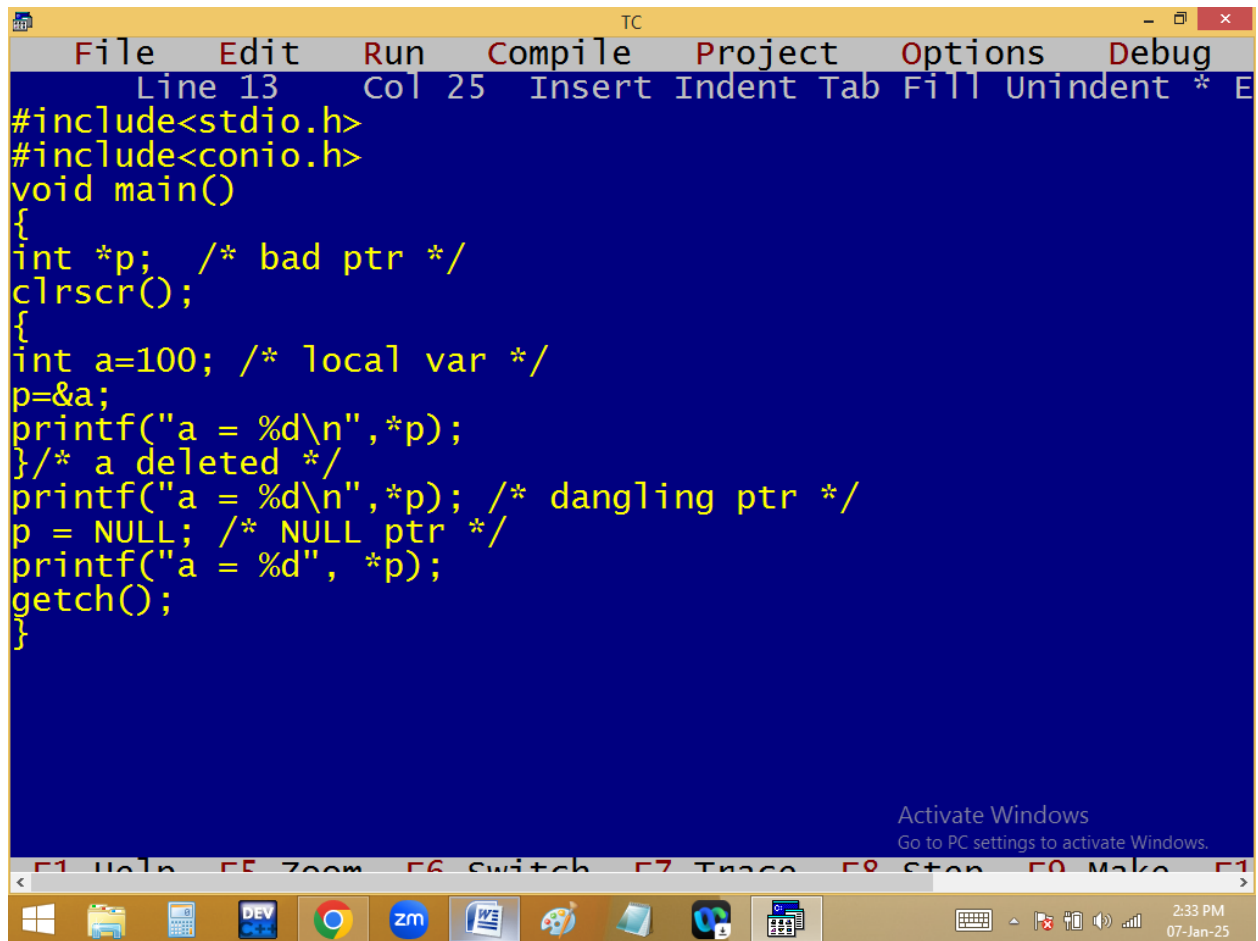
```
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* bad ptr */
clrscr();
{
int a=100; /* local var */
p=&a;
printf("a = %d\n",*p);
}/* a deleted */
printf("a=%d\n",a);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows icons: File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, VS Code, Task Manager

System tray: 2:27 PM, 07-Jan-25

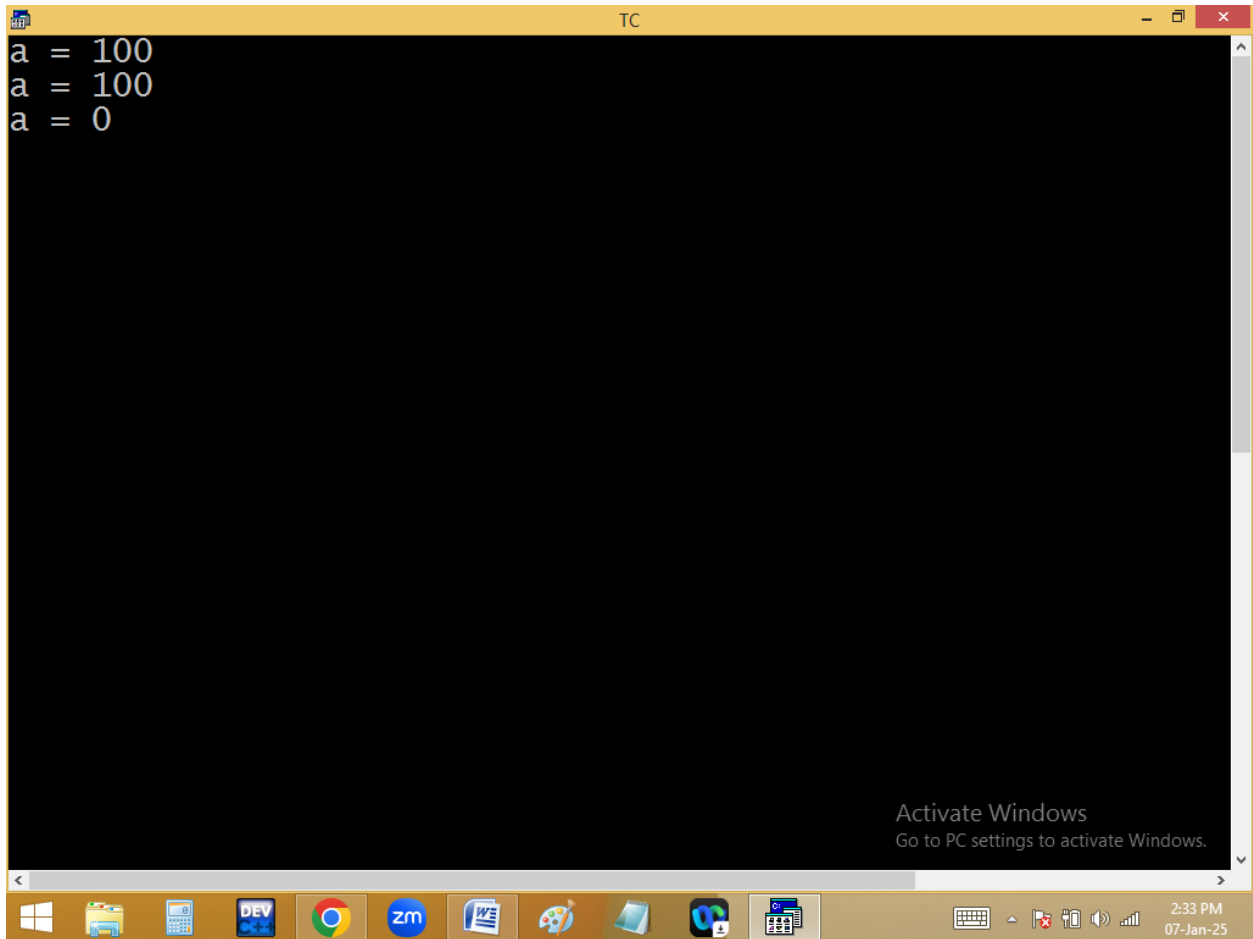


```
TC
File Edit Run Compile Project Options Debug
Line 13 Col 25 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int *p; /* bad ptr */
clrscr();
{
int a=100; /* local var */
p=&a;
printf("a = %d\n",*p);
}/* a deleted */
printf("a = %d\n",*p); /* dangling ptr */
p = NULL; /* NULL ptr */
printf("a = %d", *p);
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

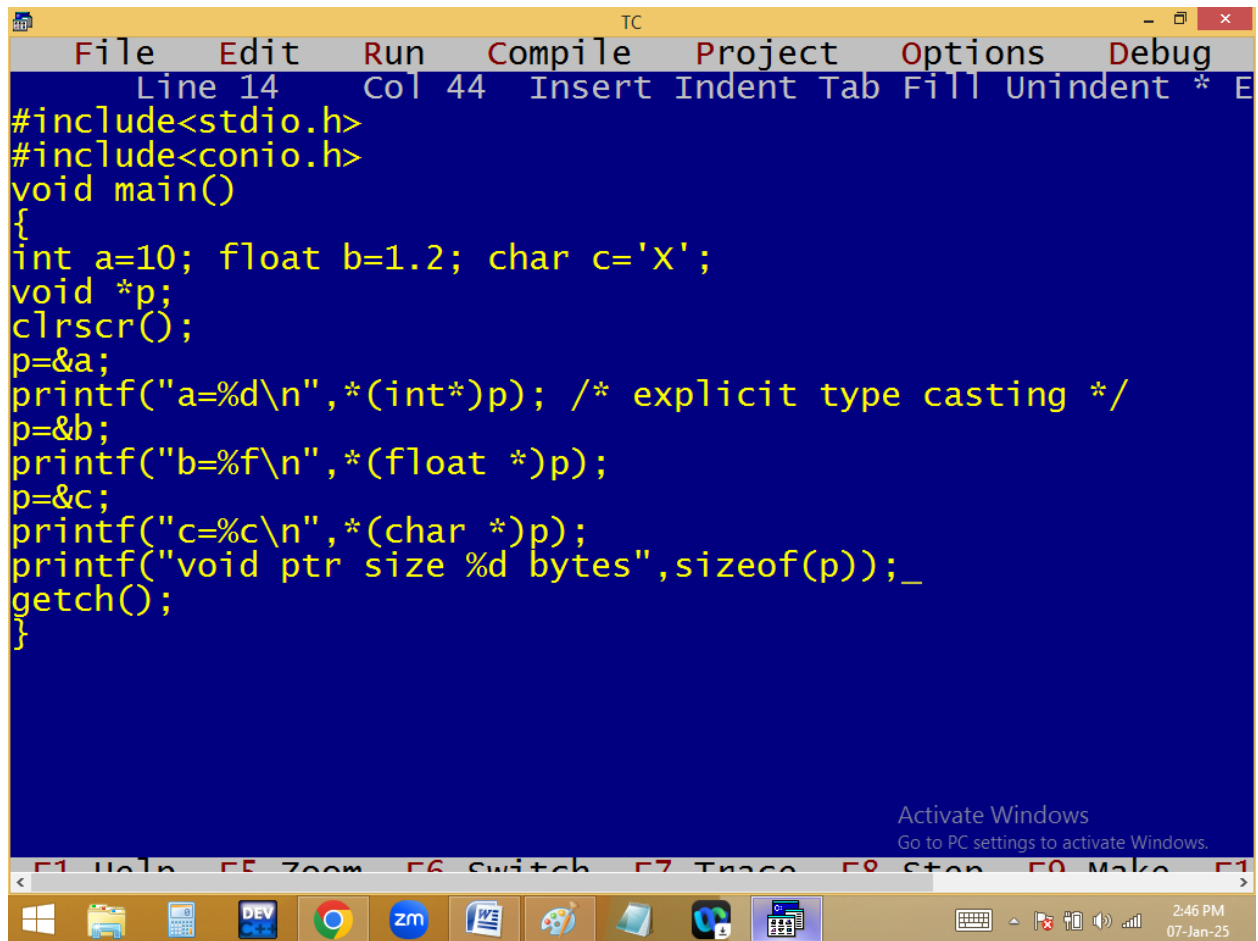
2:33 PM
07-Jan-25



```
a = 100
a = 100
a = 0
```

void / generic pointer:

pointer can store only the same type of variable address only. Using void pointer we can store any type of variable address. But before going to access the void pointer, explicit type casting should be provided. It takes 2 bytes memory and used in dynamic memory allocation.

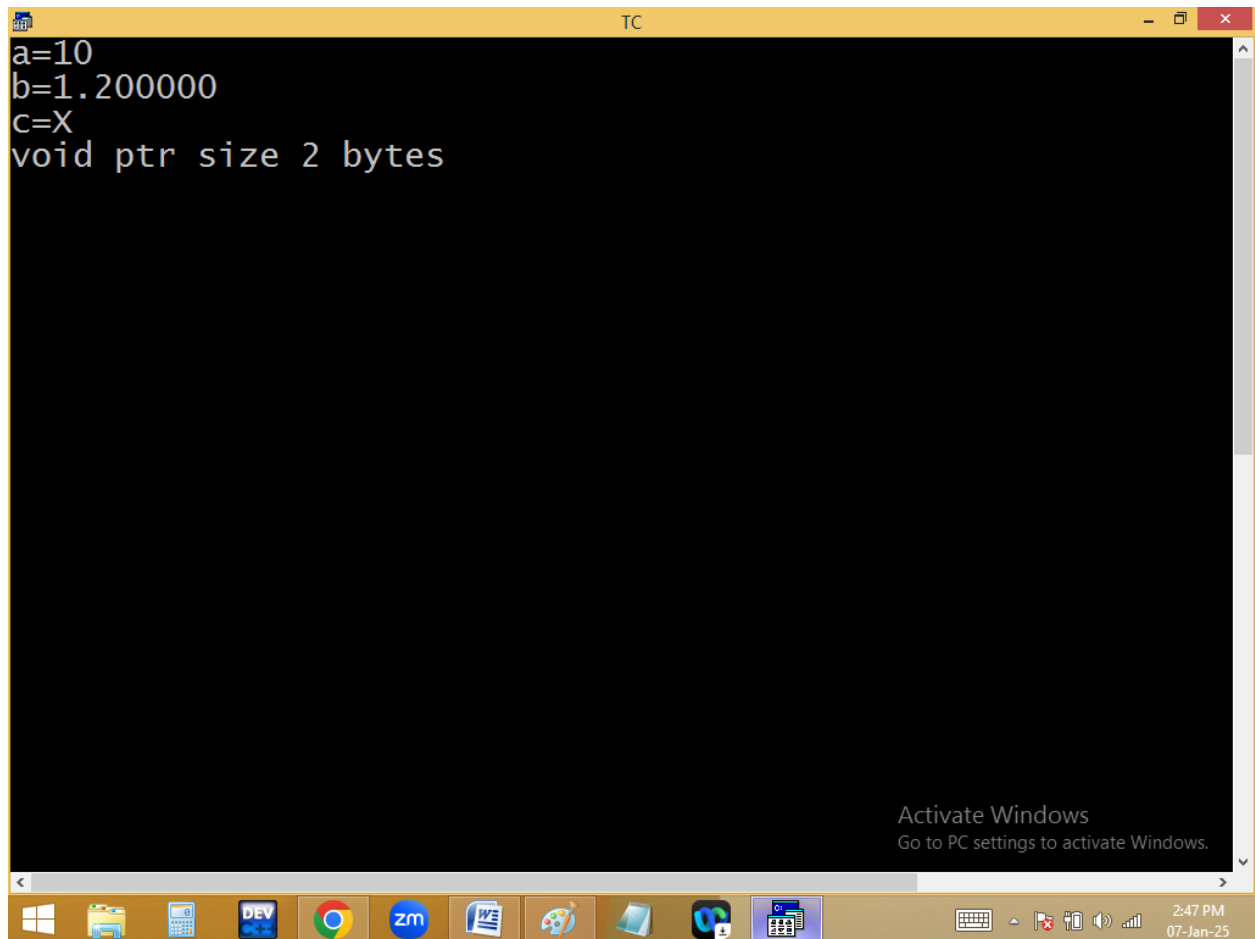


```
TC
File Edit Run Compile Project Options Debug
Line 14 Col 44 Insert Indent Tab Fill Unindent * E
#include<stdio.h>
#include<conio.h>
void main()
{
int a=10; float b=1.2; char c='X';
void *p;
clrscr();
p=&a;
printf("a=%d\n",*(int*)p); /* explicit type casting */
p=&b;
printf("b=%f\n",*(float *)p);
p=&c;
printf("c=%c\n",*(char *)p);
printf("void ptr size %d bytes",sizeof(p));_
getch();
}
```

Activate Windows
Go to PC settings to activate Windows.

F1 Help F5 Zoom F6 Switch F7 Trace F8 Stop F9 Make F10

Windows taskbar icons: File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, File Explorer, OneDrive, Task View, System tray (Keyboard, Network, Volume, Date/Time: 2:46 PM 07-Jan-25)



The image shows a screenshot of a Turbo C++ (TC) compiler window. The window has a yellow title bar with the text "TC" and standard Windows window controls (minimize, maximize, close). The main area is black with white text showing the following code:
`a=10
b=1.200000
c=X
void ptr size 2 bytes`
At the bottom right of the window, there is a message: "Activate Windows Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing various application icons and the system clock indicating 2:47 PM on 07-Jan-25.

MEMORY MANAGEMENT

To store anything in our computer, we should have to allocate the memory first.

This memory allocation is conducted in two ways.

1. Static memory allocation.
2. Dynamic memory allocation.

In static memory allocation, the memory specified at compile/design time, based on the data type or array size. This type of memory management is called compile time memory management [**compiler indicates memory and O.S allocates the memory**].

In static memory allocation, the memory size is fixed at compile time and we can't change this memory size at run time. It causes some times memory wastage / shortage.

To avoid this problem, the only solution is dynamic memory allocation.

In dynamic memory allocation, the memory is allocated at run time, based on the user input, instantly.

This type of memory management is called run time memory management.

To conduct dynamic memory allocation, we should have to use **pointers**.

In dynamic memory allocation the memory is allocated in **HEAP** area.

To manage the dynamic memory, we are using some predefined functions like

- malloc()
- calloc()
- realloc()
- free()

All these functions are available in **<alloc.h>**

malloc(), realloc(), calloc() functions are able to allocate the memory of **64KB**

Maximum at a time.

To allocate more than 64KB memory, use the functions

- `farmalloc()`
- `farcalloc()`
- `farrealloc()`.

Note:

when we are working with dynamic memory allocation, we have to allocate the memory for any data type. Due to this all these functions return datatype is **void ***, which is a generic type. Due to this we should have to provide **explicit type casting** for all these functions.

malloc()	calloc()
Block Memory allocation	Contiguous memory blocks allocation
Allocates memory in	Allocates memory in

bytes form	blocks form.
Initial values garbage	Initial values 0
One argument required	Two arguments required
Used for normal variables	Used for array type variables

Syntax:

`void * malloc(bytes);`

`void * calloc(no of blocks, block_size);`

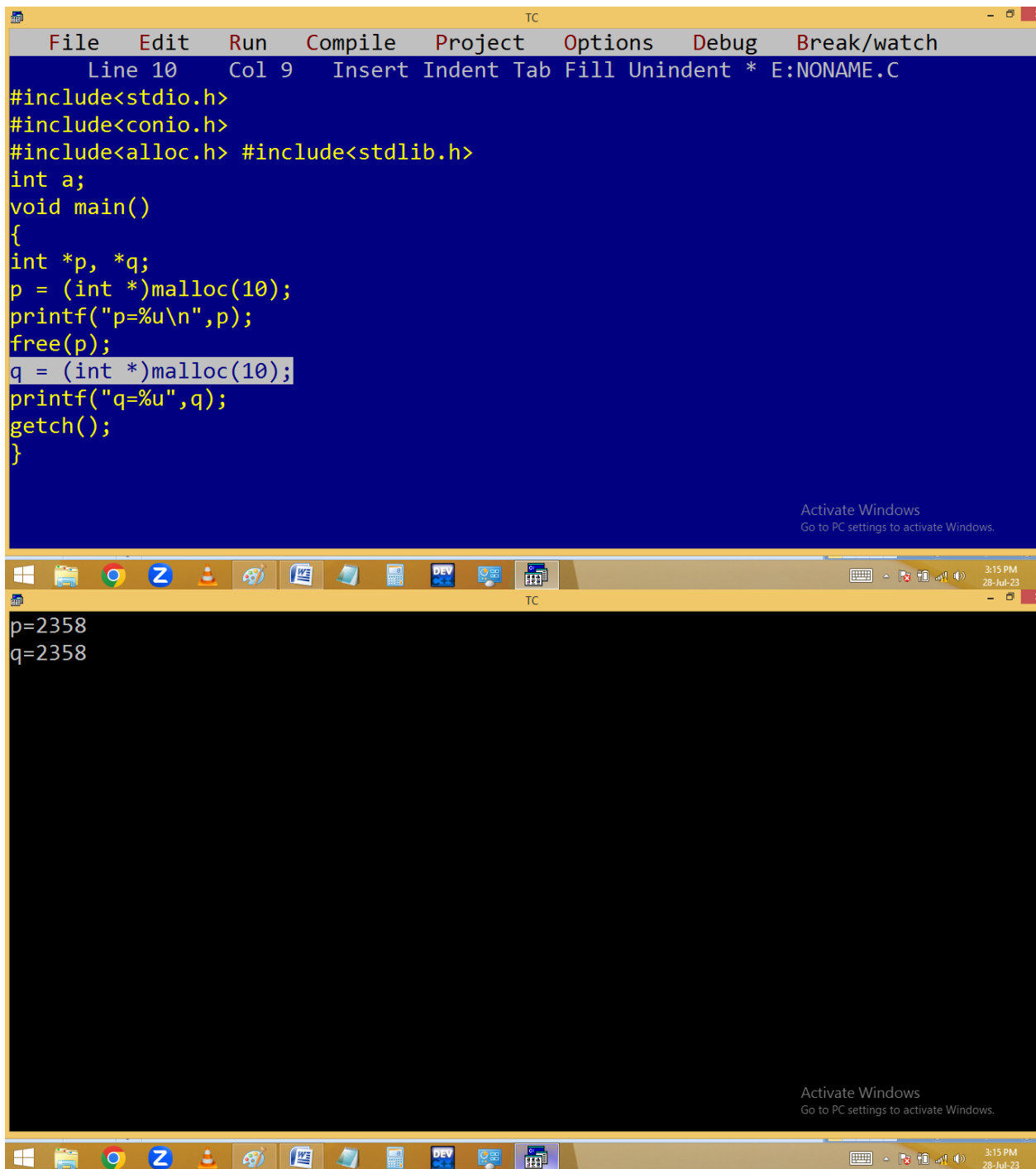
free(): It is used to release the memory allocated by `malloc()`, `calloc()` and `realloc()`.

Syntax: `void free(pointer);`

realloc(): It is used to extend the memory allocated by `malloc()` or `calloc()` at runtime. Working style is similar to `malloc()`.

Syntax: `void * realloc(oldptr, newsize);`

free example:



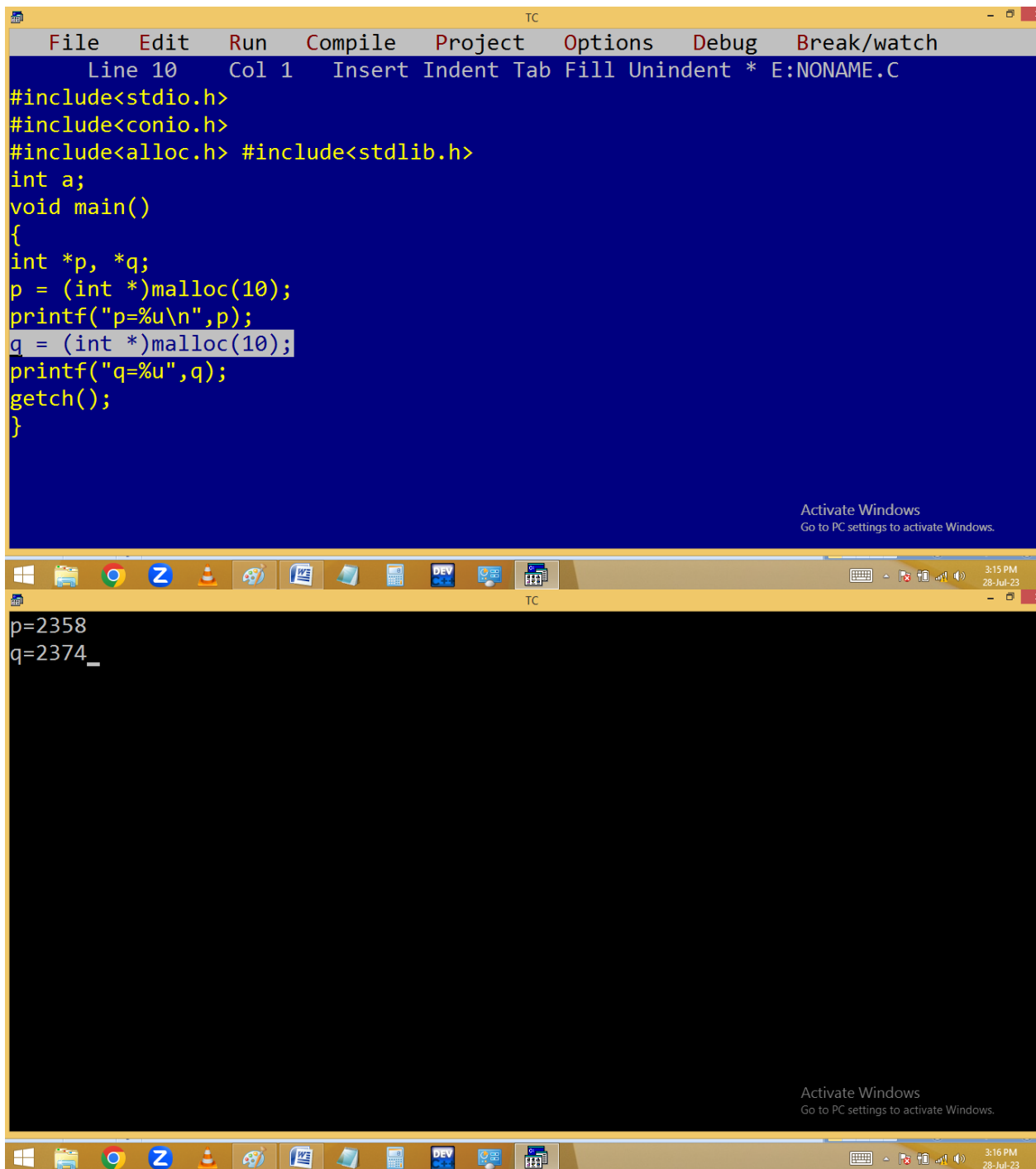
The image shows a Turbo C++ (TC) IDE window. The top menu bar includes File, Edit, Run, Compile, Project, Options, Debug, and Break/watch. The status bar at the top indicates 'Line 10 Col 9 Insert Indent Tab Fill Unindent * E:NONAME.C'. The code editor contains the following C program:

```
#include<stdio.h>
#include<conio.h>
#include<alloc.h> #include<stdlib.h>
int a;
void main()
{
    int *p, *q;
    p = (int *)malloc(10);
    printf("p=%u\n",p);
    free(p);
    q = (int *)malloc(10);
    printf("q=%u",q);
    getch();
}
```

The output window at the bottom shows the execution results:

```
p=2358
q=2358
```

Both the code editor and the output window display an 'Activate Windows' watermark in the bottom right corner, stating 'Go to PC settings to activate Windows.' The Windows taskbar at the bottom shows the time as 3:15 PM on 28-Jul-23.



The image shows a screenshot of the Turbo C++ (TC) IDE. The top window displays a C program with the following code:

```
File Edit Run Compile Project Options Debug Break/watch
Line 10 Col 1 Insert Indent Tab Fill Unindent * E:NONAME.C
#include<stdio.h>
#include<conio.h>
#include<alloc.h> #include<stdlib.h>
int a;
void main()
{
int *p, *q;
p = (int *)malloc(10);
printf("p=%u\n",p);
q = (int *)malloc(10);
printf("q=%u",q);
getch();
}
```

The bottom window shows the output of the program:

```
p=2358
q=2374_
```

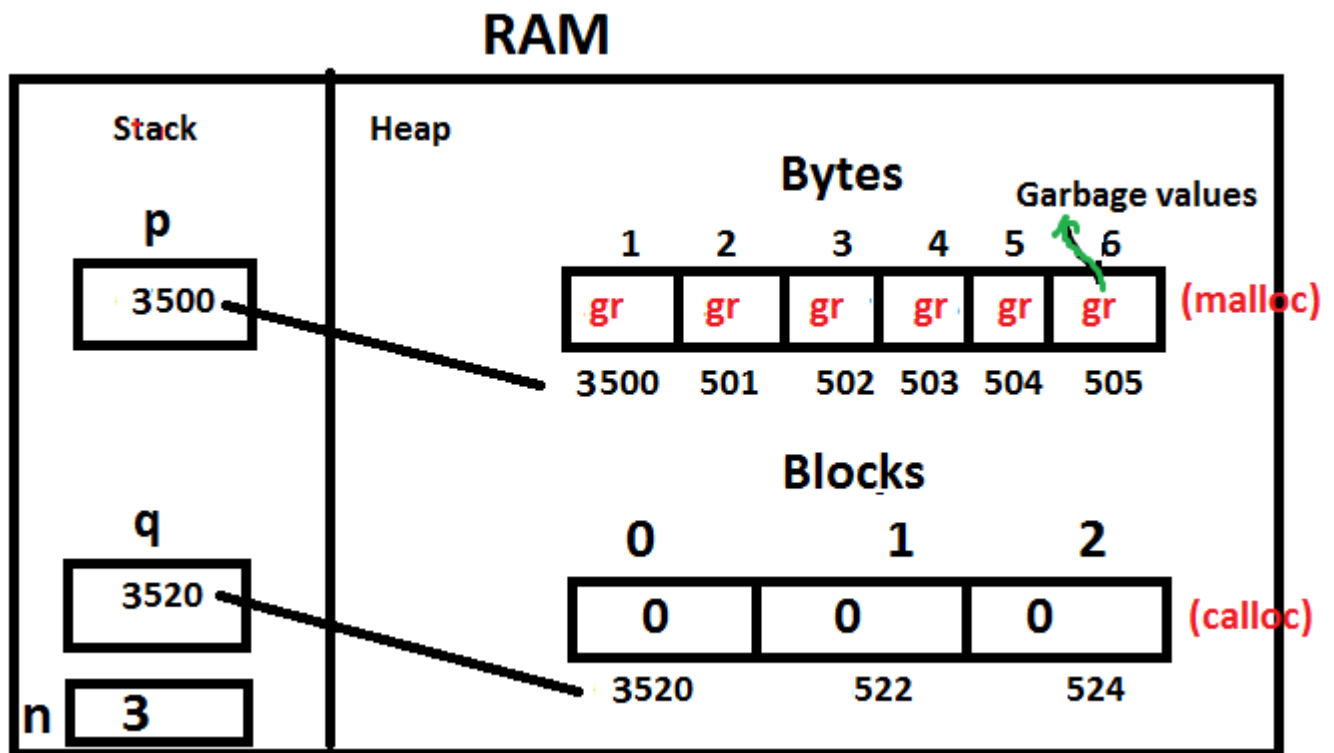
Both windows include a status bar at the bottom with the text "Activate Windows Go to PC settings to activate Windows." and a taskbar with various application icons.

allocating memory for 3 integers using malloc(), calloc().

```
int *p, *q, n=3;
```

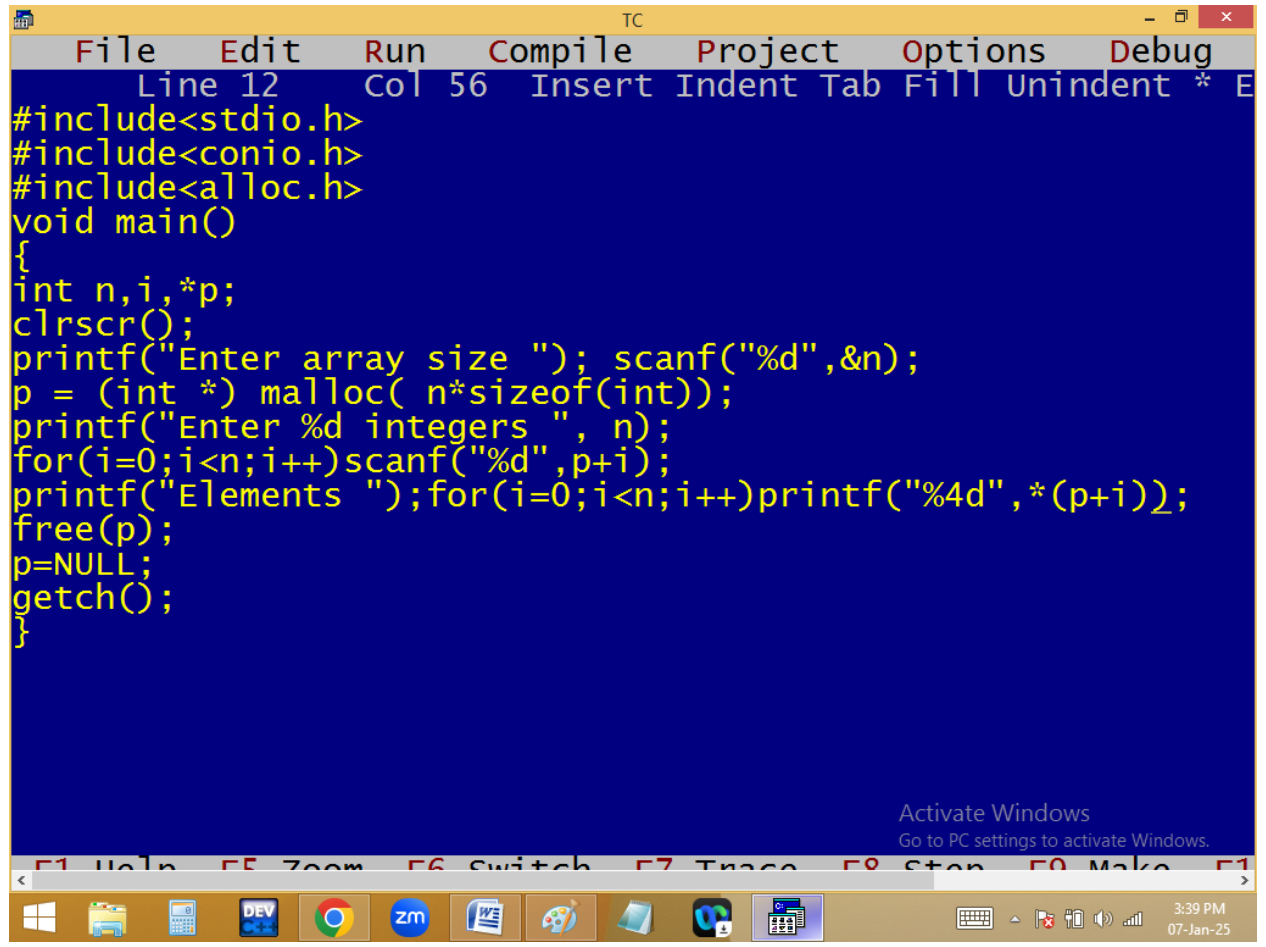
```
p = (int *)malloc(n * sizeof(int));
```

```
q = (int *)calloc(n , sizeof(int));
```



Eg:

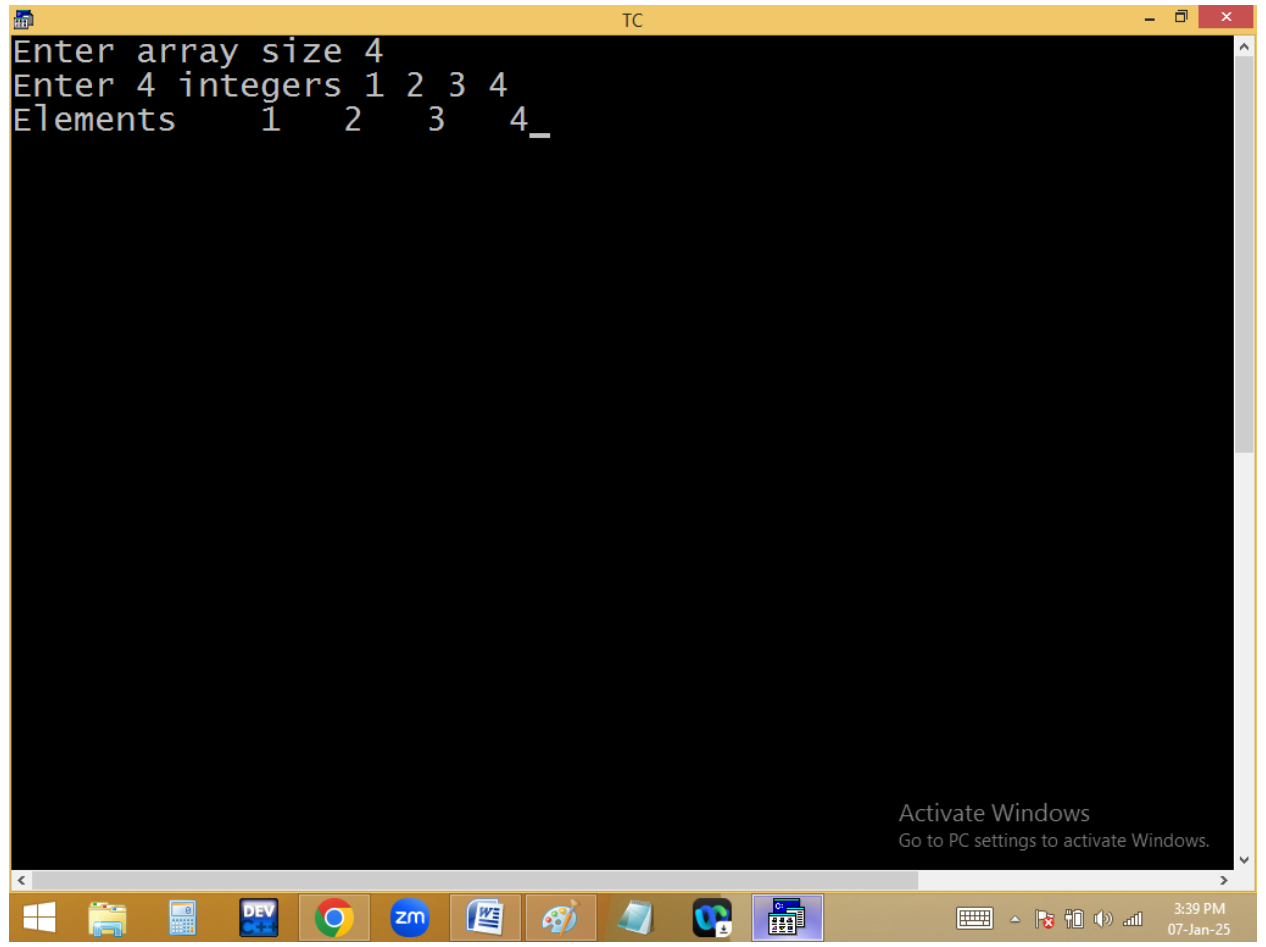
Creating dynamic one-dimensional array:

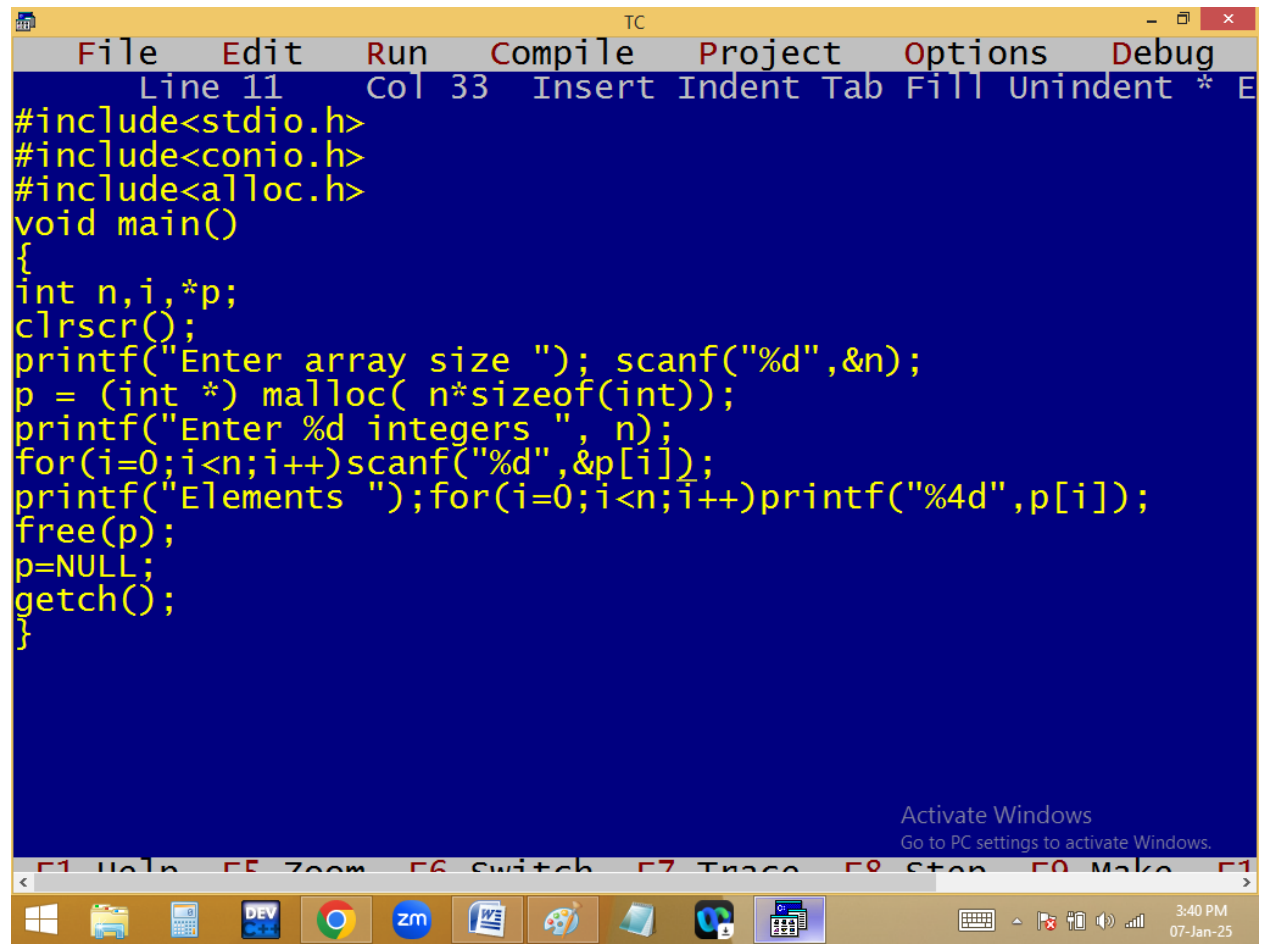


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 12", "Col 56", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#include<alloc.h>
void main()
{
int n,i,*p;
clrscr();
printf("Enter array size "); scanf("%d",&n);
p = (int *) malloc( n*sizeof(int));
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",p+i);
printf("Elements ");for(i=0;i<n;i++)printf("%4d",*(p+i));
free(p);
p=NULL;
getch();
}
```

At the bottom right of the window, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." Below the code editor is a toolbar with icons for various functions, and a taskbar at the very bottom showing icons for Windows, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, and other applications. The system clock in the bottom right corner shows "3:39 PM" and "07-Jan-25".

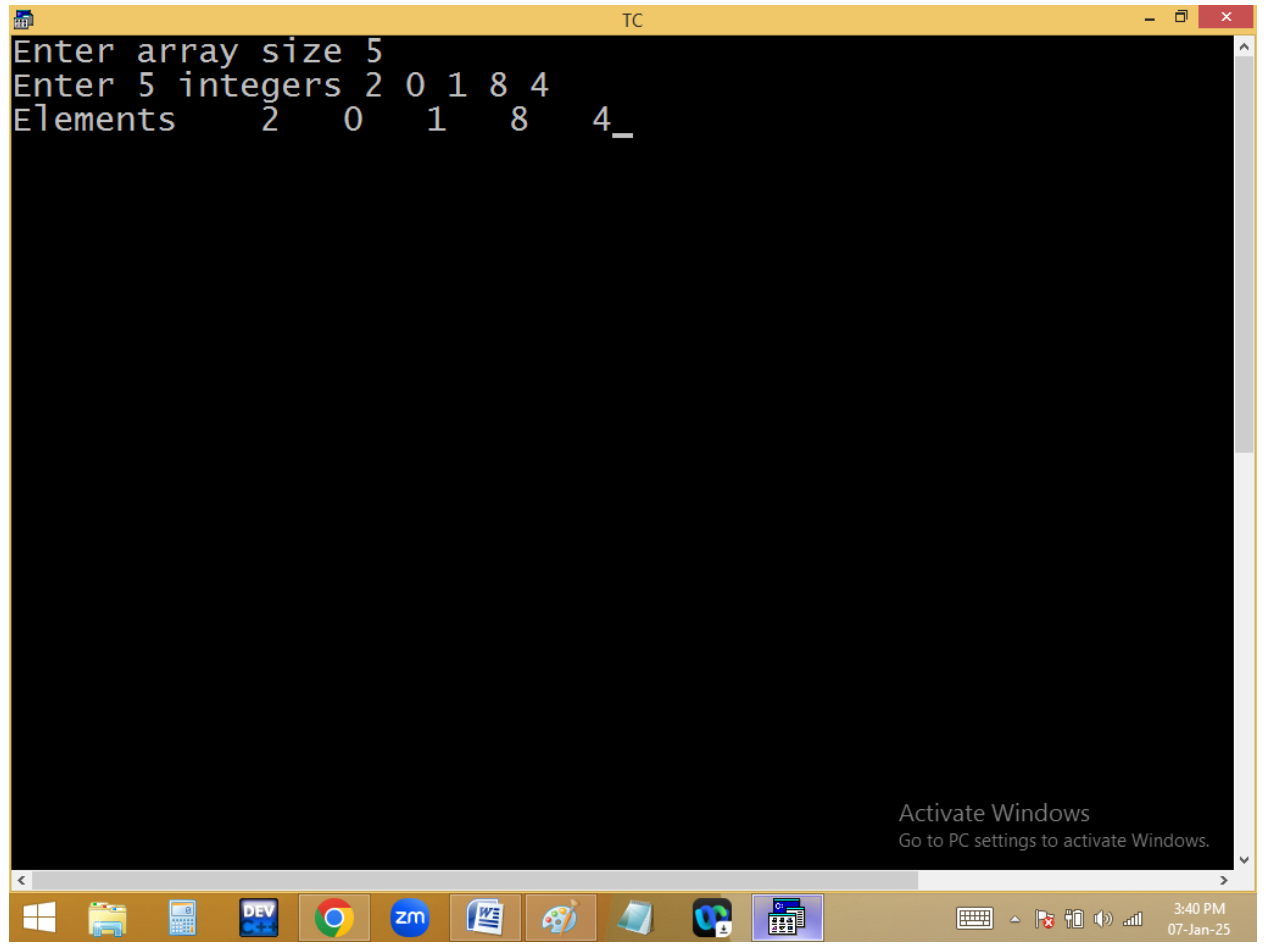


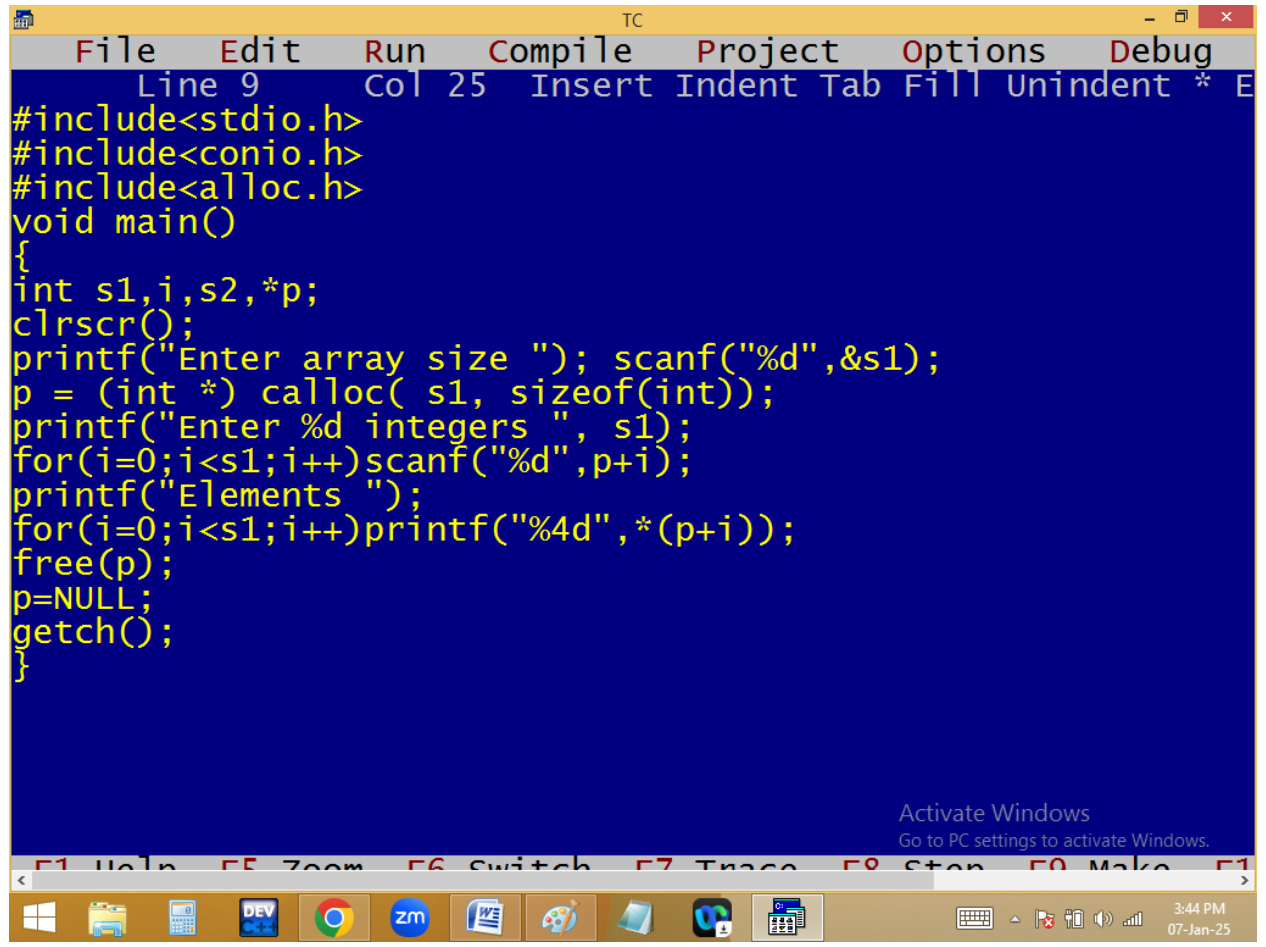


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 11", "Col 33", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#include<alloc.h>
void main()
{
int n,i,*p;
clrscr();
printf("Enter array size "); scanf("%d",&n);
p = (int *) malloc( n*sizeof(int));
printf("Enter %d integers ", n);
for(i=0;i<n;i++)scanf("%d",&p[i]);
printf("Elements ");for(i=0;i<n;i++)printf("%4d",p[i]);
free(p);
p=NULL;
getch();
}
```

At the bottom of the window, there is a Windows taskbar with icons for Windows, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and a folder icon. The system tray on the right shows the time "3:40 PM" and the date "07-Jan-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.



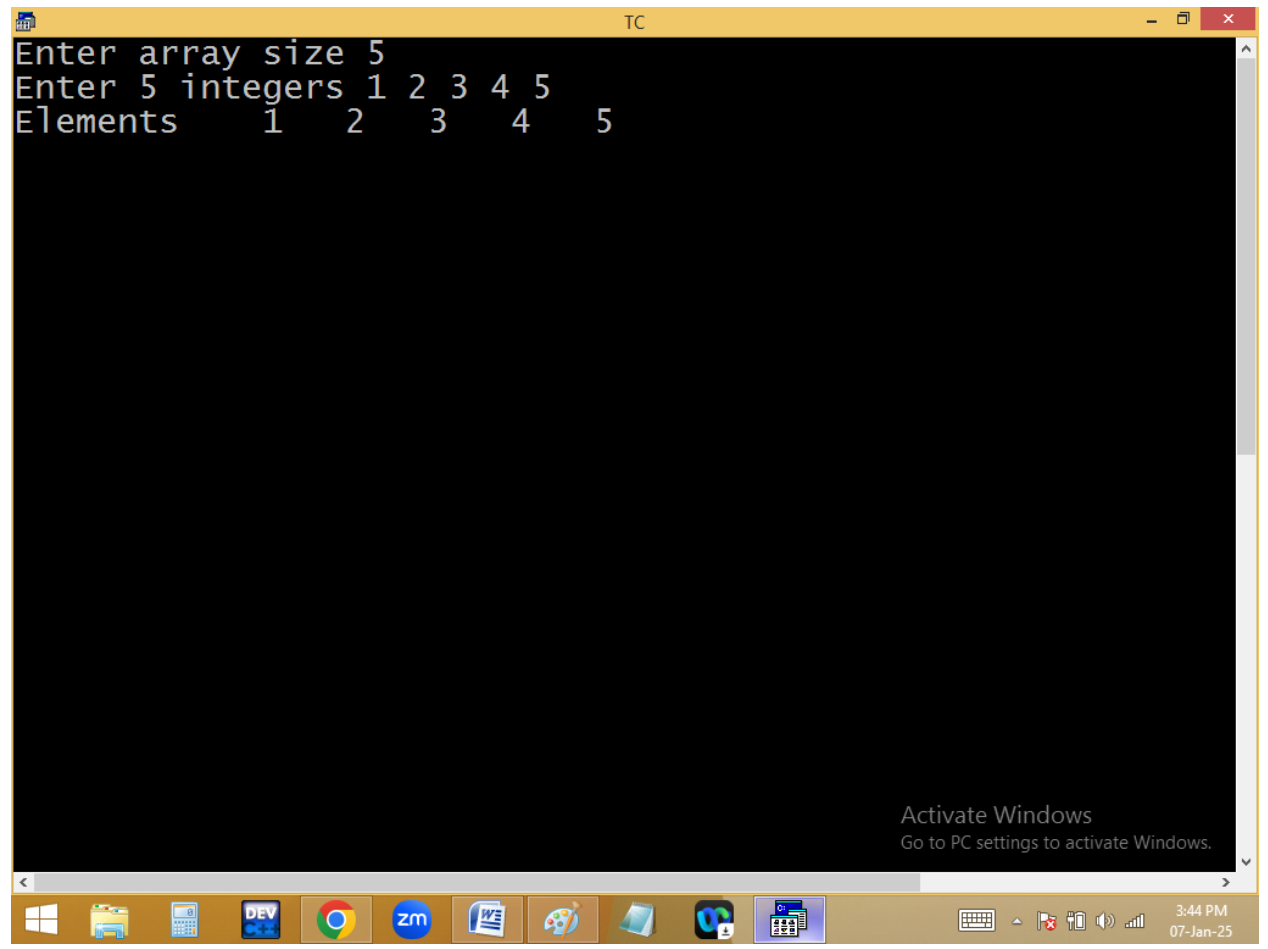


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 9" and "Col 25". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#include<alloc.h>
void main()
{
int s1,i,s2,*p;
clrscr();
printf("Enter array size "); scanf("%d",&s1);
p = (int *) calloc( s1, sizeof(int));
printf("Enter %d integers ", s1);
for(i=0;i<s1;i++)scanf("%d",p+i);
printf("Elements ");
for(i=0;i<s1;i++)printf("%4d",*(p+i));
free(p);
p=NULL;
getch();
}
```

At the bottom of the IDE, there is a toolbar with icons for various functions. A watermark "Activate Windows" is visible in the bottom right corner of the IDE window.

Below the IDE window, the Windows taskbar is visible, showing icons for the Start menu, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and a folder icon. The system tray on the right shows the time as 3:44 PM and the date as 07-Jan-25.

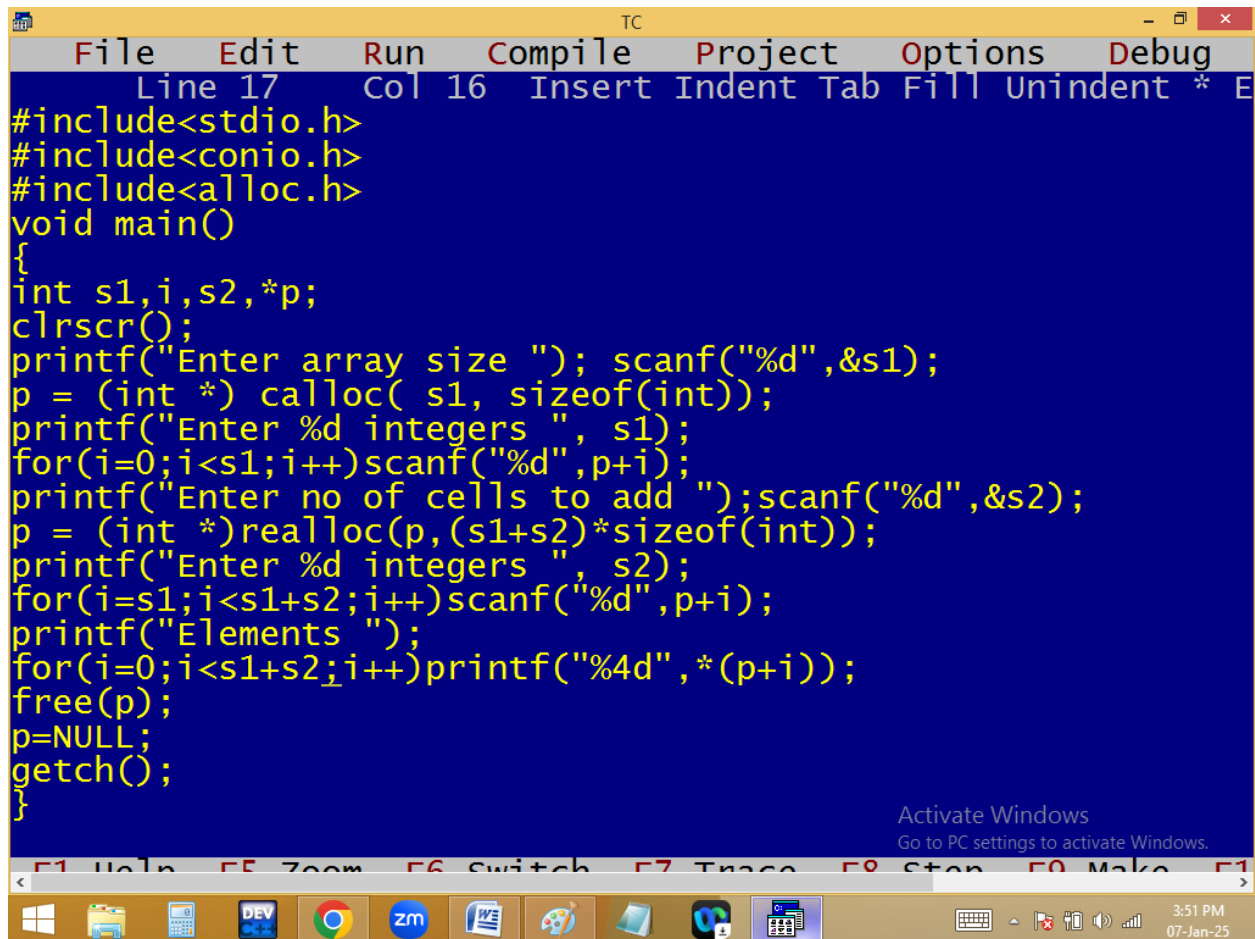


The screenshot shows a Turbo C++ (TC) window with a black background and white text. The text displays the execution of a C program that reads an array size and 5 integers, then prints them. The window title is 'TC'. The taskbar at the bottom shows various application icons and the system clock indicating 3:44 PM on 07-Jan-25.

```
Enter array size 5
Enter 5 integers 1 2 3 4 5
Elements      1      2      3      4      5
```

Activate Windows
Go to PC settings to activate Windows.

realloc() example:



The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 17", "Col 16", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "*", and "E". The main editing area has a dark blue background with yellow text. It contains a C program that merges two arrays. The code includes headers for `stdio.h`, `conio.h`, and `alloc.h`. The `main` function starts with `clrscr()`, prompts the user for an array size `s1`, allocates memory `p` using `calloc`, and reads `s1` integers. Then, it prompts for the number of cells to add `s2`, reallocates memory `p` using `realloc` to accommodate `s1+s2` integers, reads `s2` more integers, and finally prints all elements using `printf` with a width of 4. The program ends with `free(p)`, `p=NULL`, `getch()`, and a closing brace. At the bottom right of the IDE, there is a watermark that says "Activate Windows" and "Go to PC settings to activate Windows." The Windows taskbar is visible at the very bottom, showing icons for Windows, File Explorer, Calculator, DEV C++, Google Chrome, Zoom, Word, Paint, and a folder. The system clock in the bottom right corner shows "3:51 PM" and "07-Jan-25".

```
#include<stdio.h>
#include<conio.h>
#include<alloc.h>
void main()
{
int s1,i,s2,*p;
clrscr();
printf("Enter array size "); scanf("%d",&s1);
p = (int *) calloc( s1, sizeof(int));
printf("Enter %d integers ", s1);
for(i=0;i<s1;i++)scanf("%d",p+i);
printf("Enter no of cells to add ");scanf("%d",&s2);
p = (int *)realloc(p,(s1+s2)*sizeof(int));
printf("Enter %d integers ", s2);
for(i=s1;i<s1+s2;i++)scanf("%d",p+i);
printf("Elements ");
for(i=0;i<s1+s2;i++)printf("%4d",*(p+i));
free(p);
p=NULL;
getch();
}
```

```
TC
Enter array size 3
Enter 3 integers 1 0 3
Enter no of cells to add 2
Enter 2 integers 5 9
Elements      1    0    3    5    9
```

Activate Windows
Go to PC settings to activate Windows.

